

DVIGRAD STARGAZING PARTY — MASTER FIELD GUIDE

Date: Saturday, September 12, 2026 | **Observing Window:** 20:00 – 22:30 CEST (UTC+2)
Location: Dvigrad Fortress Ruins, Kanfanar, Istria, Croatia (45.12637° N, 13.81296° E, Elev 143 m)
Instrument: Sky-Watcher Skyliner Classic 200P Dobsonian (200 mm / 8" Newtonian Reflector, f/6, 1200 mm FL)

LOCATION & LIGHT POLLUTION PROFILE

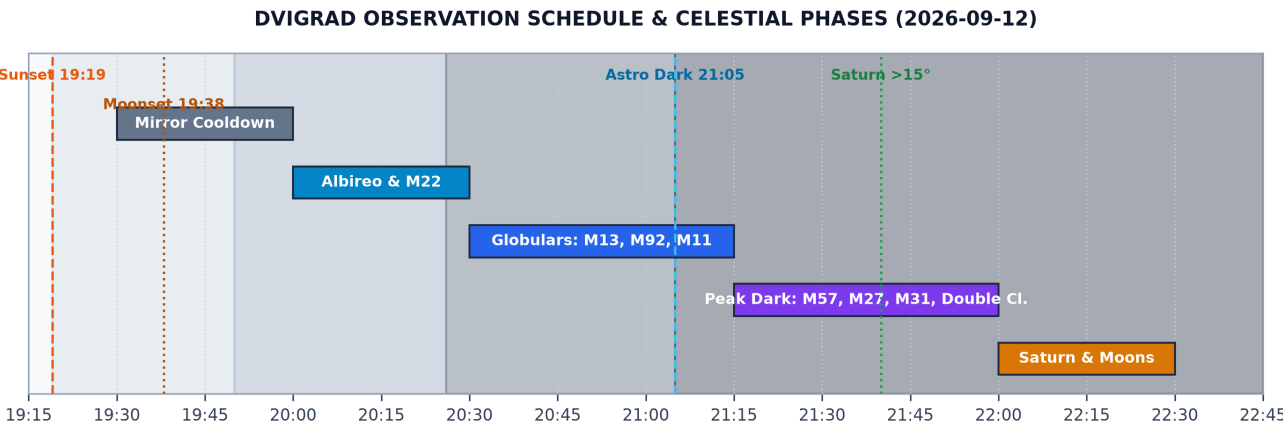
- Bortle Class:** Class 4 (Rural / Suburban transition)
- Zenith Sky Brightness:** 20.86 – 21.15 mag/arcsec² (SQM)
- Naked-Eye Limiting Mag (NELM):** 6.0 – 6.2 mag overhead
- Valley Topography:** Deep Draga depression blocks coastal light domes (Rovinj & Pula). Limiting tree & terrain obstruction: 10°–15° cutoff.

TELESCOPE & OPTICAL CONFIGURATION

- 200P Dobsonian:** 200 mm aperture, f/6, 1200 mm focal length (820× eye)
- 20 mm Eyepiece (Finder/Wide):** 60× mag | 3.3 mm exit pupil | 50' (0.83°) TFOV
- 12.5 mm Eyepiece (High Power):** 96× mag | 2.1 mm exit pupil | 31' (0.52°) TFOV
- Orientation:** Newtonian inverted 180° (up=down, left=right). RACI=upright.

MOON EPHEMERIS & DARK SKY VERDICT: ZERO LUNAR LIGHT POLLUTION
On Sep 12, 2026, the Moon is a razor-thin **3.1% waxing crescent** (1.1 days after New Moon) and **sets at 19:38 CEST** (prior to start). Throughout the observing party (20:00–22:30 CEST), the Moon is **below the horizon (-4° to -30° Alt)**. Pristine Bortle 4 contrast for faint deep-sky nebulae and galaxies!

ASTRONOMICAL TIMELINE & OBSERVATION SCHEDULE



Phase	Time Window (CEST)	Sun Alt	Target Program & Operational Guidance
Setup & Cooldown	19:30 – 20:00	-2° to -8°	Primary mirror cooldown (30-45 min). Finder alignment on Vega / Arcturus.
Nautical Twilight	20:00 – 20:26	-8° to -12°	Bright double star Albireo . Urgent early sweep for M22 before southern trees block it.
Astro Twilight	20:26 – 21:05	-12° to -18°	Sky darkens, Milky Way Great Rift emerges. Hercules globulars M13 and M92 .
True Astro Darkness	21:05 – 22:30+	< -18.0°	Zero solar glow. Peak deep-sky window: M57, M27, M31 / M32 / M110, Double Cluster.
Planetary Finale	22:00 – 22:30	< -24°	Saturn rises above 20° in ESE. Edge-on ring needle geometry, Titan & icy moons.

CURATED TARGET CATALOG (MID-SESSION EPHEMERIS @ 21:15 CEST)

Targets selected for high visual impact in 200 mm aperture, respecting the 10°–15° Dvigrad horizon obstruction. Step-by-step star-hopping instructions and inverted eyepiece simulations are on the referenced charts.

#	Target Name	Const	Type	Mag	Size	Difficulty	Chart Ref	Recommended Eyepiece
1	Albireo (β Cyg)	Cyg	Double Star	3.1/5.1	34.3"	Very Easy	Chart 2	20mm / 12.5mm
2	M22 (NGC 6656)	Sgr	Globular Cl.	5.1	32.0'	Easy	Full Sky	20mm → 12.5mm
3	M13 (Great Globular)	Her	Globular Cl.	5.8	20.0'	Easy	Chart 3	12.5mm (96×)
4	M92 (NGC 6341)	Her	Globular Cl.	6.3	14.0'	Easy	Chart 3	12.5mm (96×)
5	M11 (Wild Duck)	Sct	Open Cluster	5.8	14.0'	Easy	Chart 6	20mm & 12.5mm
6	M57 (Ring Nebula)	Lyr	Planetary Neb.	8.8	1.4'×1.0'	Easy	Chart 1	12.5mm (96×)
7	M27 (Dumbbell)	Vul	Planetary Neb.	7.4	8.0'×5.6'	Medium	Chart 2	20mm (60×)
8	Double Cluster	Per	Dual Open Cl.	3.7/3.8	60.0'	Easy	Chart 5	20mm (60×)
9	M31 (Andromeda)	And	Spiral Galaxy	3.4	190'×60'	Easy	Chart 4	20mm (60×)
10	Saturn & Titan	Aqr	Planet & Rings	+0.6	19.2"	Very Easy	Chart 6	12.5mm (96×)

PRACTICAL FIELD PROTOCOLS FOR DVIGRAD RUINS

FIELD PROTOCOLS & OBSERVING TIPS

• **Primary Mirror Acclimation:** Set up the 200P Dobsonian by 19:30 CEST. An 8" parabolic mirror requires 30–45 min to reach thermal equilibrium. Tube air currents degrade high-power views of M13, M57, and Saturn until fully cooled.

• **Valley Dew Management:** The Draga depression concentrates cool, damp air. Keep unmounted eyepieces in warm jacket pockets or closed cases. Replace the front dust cover on the telescope when taking socializing breaks.

• **Tracking Near Zenith ('Dobson Hole'):** Objects straight overhead (>80° Alt, e.g. M57 & Albireo) are stiff to track in an Alt-Az mount. Observe M57 between 21:15 and 21:45 as it rolls west of the meridian (70°–75° Alt) for effortless manual nudging.

• **Red Light & Dark Adaptation:** Full retinal rod-cell dark adaptation requires 20–30 min. Strict deep red lights only—avoid phone screens to preserve Bortle 4 contrast on faint nebula lobes and galactic dust lanes.

• **Saturn Elevation Timing:** Saturn rises around 20:20 CEST but clears the 15° Dvigrad tree line at 21:40. Reserve 22:00–22:30 for the planetary finale at >20° altitude for stable seeing.

PRINTABLE SKY CHARTS & MAPS DIRECTORY (TONER-SAVER NEGATIVE)

Chart / Map Document	Target Objects & Region	Eyepiece Simulation & FOV	File Path
Full Sky Map	All constellations, Milky Way, 15° obstruction ring	All-sky naked-eye planisphere	observations/dvigrad-2026-09-12/full_sky_map.pdf
Finder Chart 1	M57 Ring Nebula in Lyra	12.5mm (96×, 31' FOV) Inverted Negative	charts/chart_1_lyra_m57.pdf
Finder Chart 2	M27 Dumbbell Nebula & Albireo Double	20mm (60×, 50' FOV) Inverted Negative	charts/chart_2_vulpecula_m27_albir eo.pdf
Finder Chart 3	Hercules Globulars M13 & M92	12.5mm (96×, 31' FOV) Inverted Negative	charts/chart_3_hercules_m13_m92. pdf
Finder Chart 4	M31 Andromeda Galaxy, M32 & M110	20mm (60×, 50' FOV) Inverted Negative	charts/chart_4_andromeda_m31.pdf
Finder Chart 5	Perseus Double Cluster (NGC 869 / 884)	20mm (60×, 50' FOV) Inverted Negative	charts/chart_5_perseus_double_clu ster.pdf
Finder Chart 6	M11 Wild Duck Cluster & Saturn Ring View	Dual Inset: 12.5mm M11 & Saturn	charts/chart_6_scutum_m11_and_s aturn.pdf